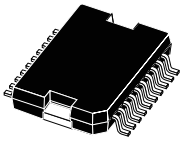
L6234

Three-phase motor driver

Datasheet - production data



PowerSO20

Features

- Supply voltage from 7 to 52 V
- 5 A peak current
- R_{DSon} 0.3 Ω typ. value at 25 °C
- Cross conduction protection
- TTL compatible driver
- Operating frequency up to 150 kHz
- Thermal shutdown
- Intrinsic fast free wheeling diodes
- Input and enable function for each half bridge
- 10 V external reference available

Description

The L6234 is a triple half bridge to drive a brushless DC motor.

It is realized in BCDmultipower technology which combines isolated DMOS power transistors with CMOS and bipolar circuits on the same chip.

By using mixed technology it has been possible to optimize the logic circuitry and the power stage to achieve the best possible performance.

The output DMOS transistors can sustain a very high current due to the fact that the DMOS structure is not affected by the second breakdown effect, the RMS maximum current is practically limited by the dissipation capability of the package.

All the logic inputs are TTL, CMOS and µP compatible. Each channel is controlled by two separate logic inputs.

The L6234 device is available in a 20-pin PowerSO20 package.

Table 1. Device summary

Order code	Package	Packing
L6234PD	PowerSO20	Tube
L6234PD013TR	PowerSO20	Tape and reel

This is information on a product in full production. www.st.com

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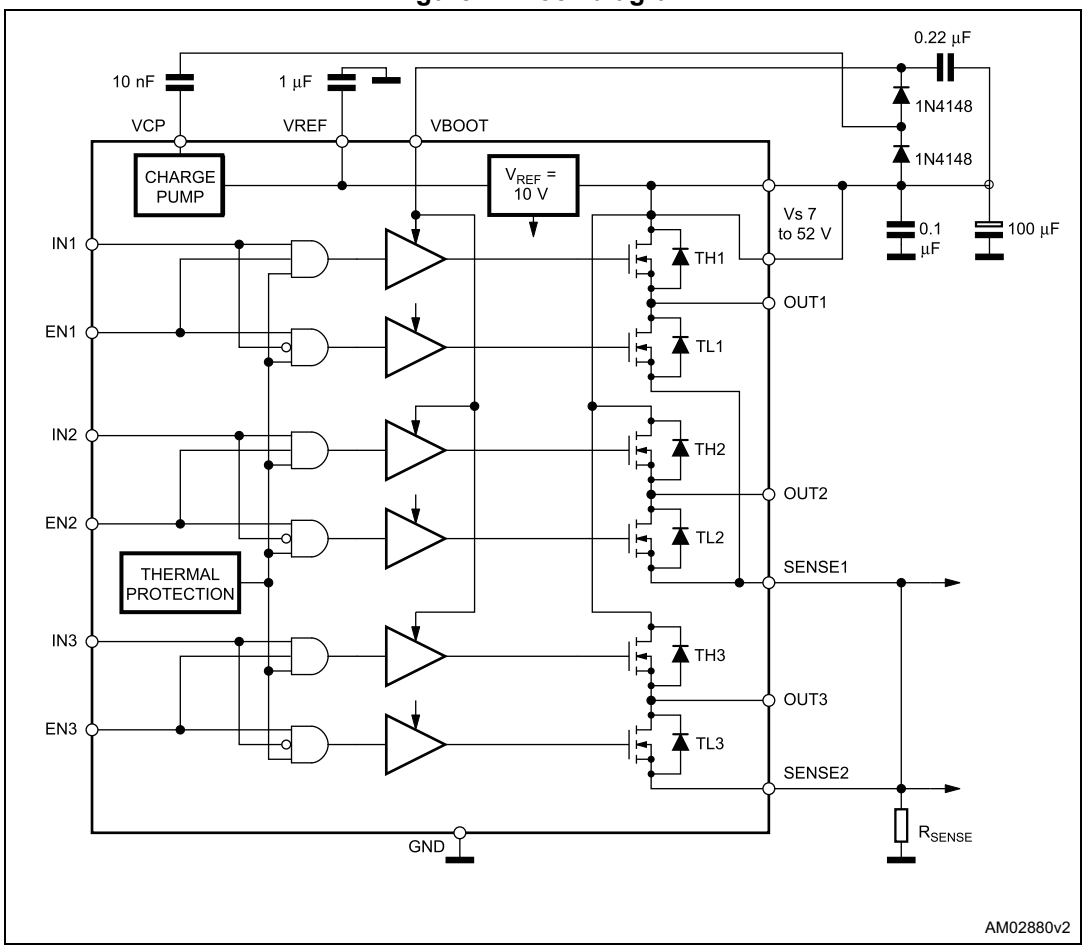
1 Block diagram

Figure 1. Block diagram



2 Pin connections

Figure 2. Pin connections

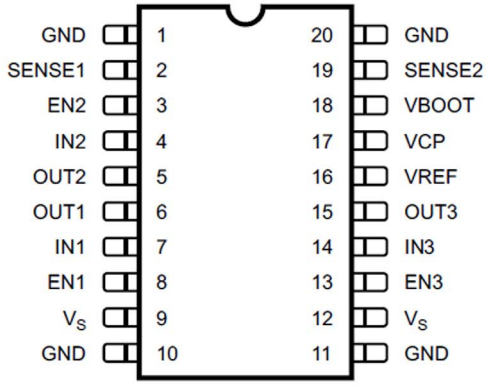


Table 2. Pin functions

Pin no.	Name	Function
1, 10 11, 20	GND	Common ground terminal. In the PowerSO package these pins are used to dissipate the heat forward the PCB.
2	SENSE1	A sense resistor connected to this pin provides feedback for motor current control for the bridges 1 and 2.
3 8 13	EN 2 EN 1 EN 3	Enable of the channels 1/2/3. A logic LOW level on this pin switches off both power DMOS of the related channel.
4 7 14	IN 2 IN 1 IN 3	Logic input of channels 1/2/3. A logic HIGH level (when the corresponding EN pin is HIGH) switches ON the upper DMOS power transistor, while a logic LOW switches ON the corresponding low side power DMOS.
5 6 15	OUT 2 OUT 1 OUT 3	Output of the channels 1/2/3.
9, 12	V_S	Power supply voltage.
16	VREF	Internal voltage reference. A capacitor connected from this pin to GND increases the stability of the power DMOS drive circuit.
17	VCP	Bootstrap oscillator. Oscillator output for the external charge pump.
18	VBOOT	Overvoltage input to drive the upper DMOS
19	SENSE2	A sense resistor connected to this pin provides feedback for motor current control for the bridge 3.

3 Thermal data

Table 3. Thermal data

Symbol	Parameter	PowerSO20	Unit
$R_{th\ j\text{-}case}$	Thermal resistance junction case	1.5	°C/W

4 Maximum ratings

Table 4. Absolute maximum ratings

Symbol	Parameter	Value	Unit
V_S	Power supply voltage	52	V
V_{IN}, V_{EN}	Input enable voltage	– 0.3 to 7	V
I_{peak}	Pulsed output current(1)	5	A
V_{SENSE}	Sensing voltage (DC voltage)	-1 to 4	V
V_{boot}	Bootstrap peak voltage	62	V
V_{OD}	Differential output voltage (between any of the 3 OUT pins)	60	V
f_C	Commutation frequency	150	kHz
V_{REF}	Reference voltage	12	V
P_{tot}	Total power dissipation (T_A = 70 °C)	2.3	W
T_{stg}, T_j	Storage and junction temperature range	-40 to 150	°C

1. Pulse width limited only by junction temperature and the transient thermal impedance.

Recommended operating conditions

Table 5. Recommended operating conditions

Symbol	Parameter	Value	Unit
V_S	Supply voltage	7 to 42	V
V_{OD}	Peak to peak differential voltage (between any of the 3 out pins)	52	V
I_{out}	DC output current (T_A = 25 °C)	4	A
V_{SENSE}	Sensing voltage (pulsed t_w < 300 nsec)	-4 to 4	V
	Sensing voltage (DC)	-1 to 1	V
T_j	Junction temperature range	-40 to 125	°C

5 Electrical characteristics

V_S = 42 V; T_j = 25 °C unless otherwise specified.

Table 6. Electrical characteristics

Symbol	Parameter	Test condition	Min.	Typ.	Max.	Unit
V_S	Supply voltage	-	7	-	52	V
V_{ref}	Reference voltage	-	-	10	-	V
I_S	Quiescent supply current	-	-	6.5	-	mA
T_S	Thermal shutdown	-	150	-	-	°C
T_D	Dead time protection	-	-	300	-	ns
Output DMOS transistor						
I_{DSS}	Leakage current	-	-	-	1	mA
R_{DSon}	ON resistance	-	-	0.3	-	Ω
Source drain diode						
V_{SD}	Forward ON voltage	I_{SD} = 4 A; EN = LOW	-	1.2	-	V
T_{RR}	Reverse recovery time	I_F = 4 A	-	900	-	ns
T_{pr}	Forward recovery time	-	-	200	-	ns
Logic levels						
V_{INL}, V_{ENL}	Input LOW voltage	-	-0.3	-	0.8	V
V_{INH}, V_{ENH}	Input HIGH voltage	-	2	-	7	V
I_{INL}, I_{ENL}	Input LOW current	V_{IN}, V_{EN} = L	-		-10	µA
I_{INH}, I_{ENH}	Input HIGH current	V_{IN}, V_{EN} = H	-	30	-	µA

6 Circuit description

The L6234 is a triple half bridge designed to drive brushless DC motors. Each half bridge has 2 power DMOS transistors with R_{DSon} = 0.3 Ω.

The 3 half bridges can be controlled independently by means of the 3 inputs IN1, IN2, IN3 and the 3 inputs EN1, EN2, and EN3. An external connection to the 3 common low side DMOS sources is provided to connect a sensing resistor for the constant current chopping application.

The driving stage and the logic stage are designed to work from 7 V to 52 V.

7 Typical characteristics

Figure 3. Quiescent current vs. supply voltage

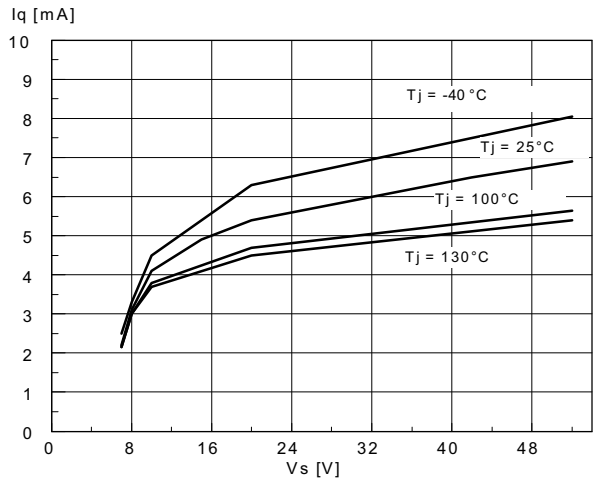


Figure 4. Normalized quiescent current vs. switching frequency

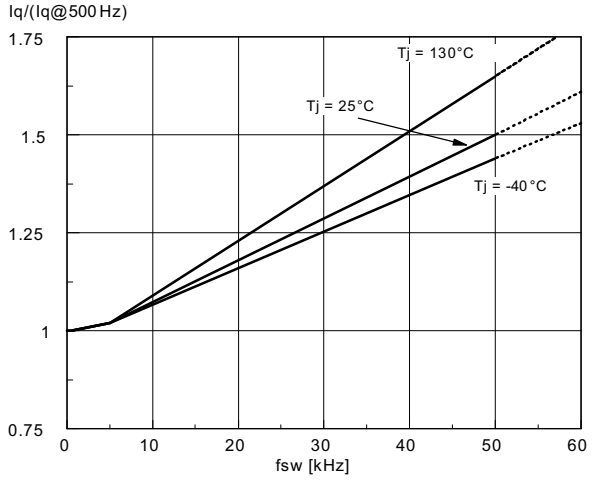


Figure 5. Typical R_{DSon} vs. supply voltage

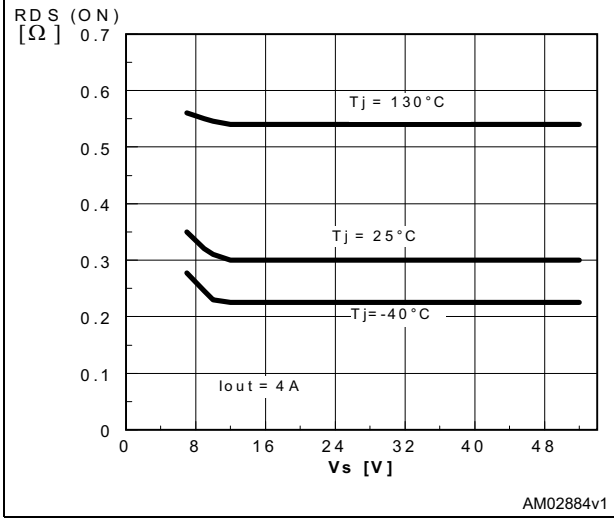


Figure 6. Source drain forward on voltage vs. junction temperature

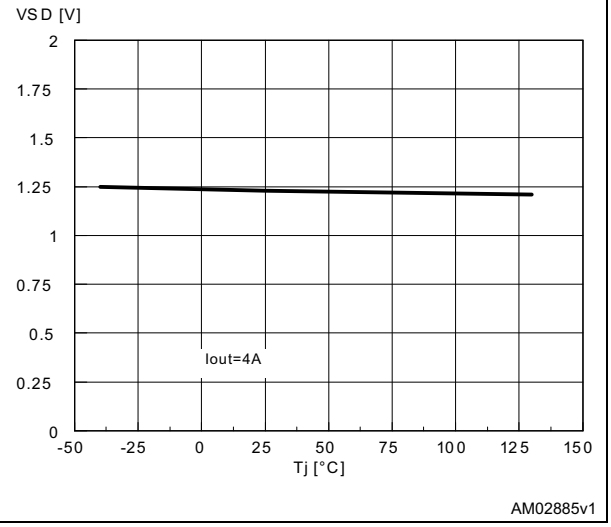


Figure 7. Typical diode forward ON characteristics

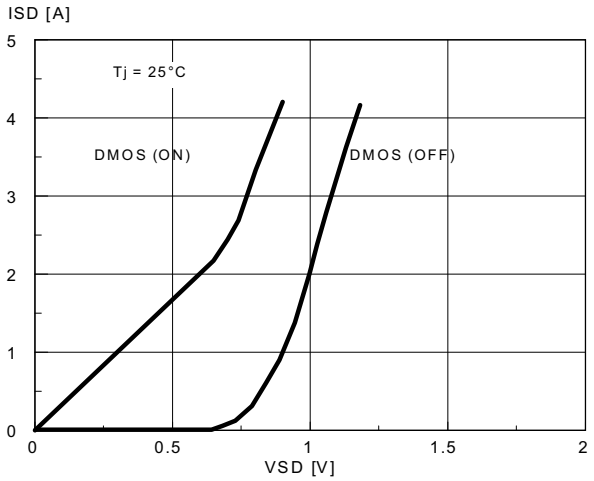


Figure 8. Reference voltage vs. supply voltage

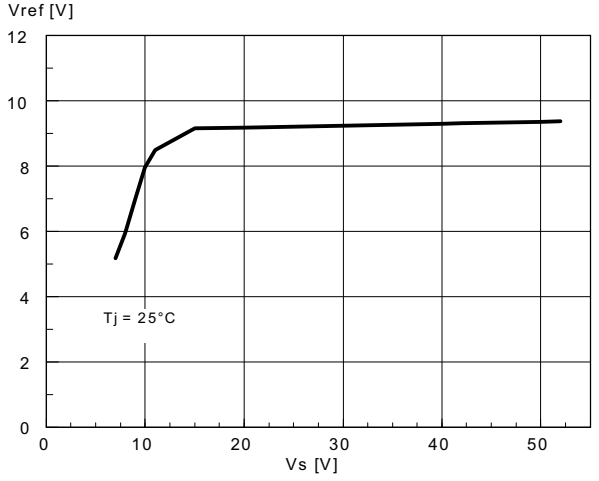


Figure 9. Reference voltage vs. junction temperature

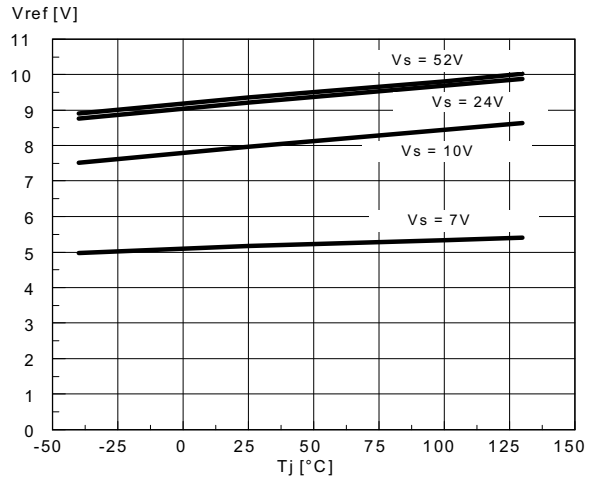


Figure 10. PowerSO20 transient thermal resistance

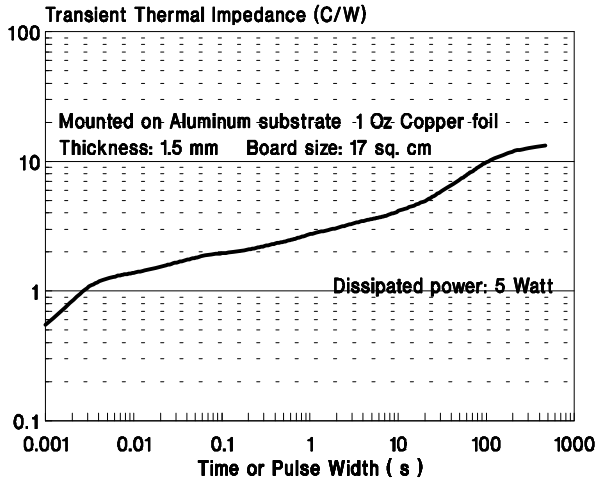


Figure 11. PowerSO20 thermal resistance (mounted on aluminium substrate)

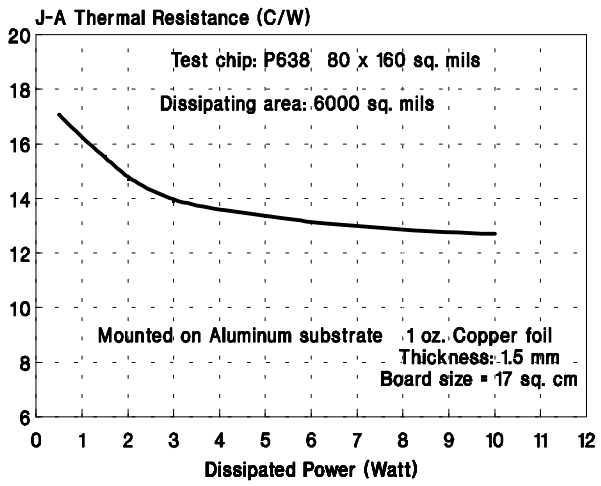


Figure 12. PowerSO20 thermal resistance (mounted on FR4 monolayer substrate)

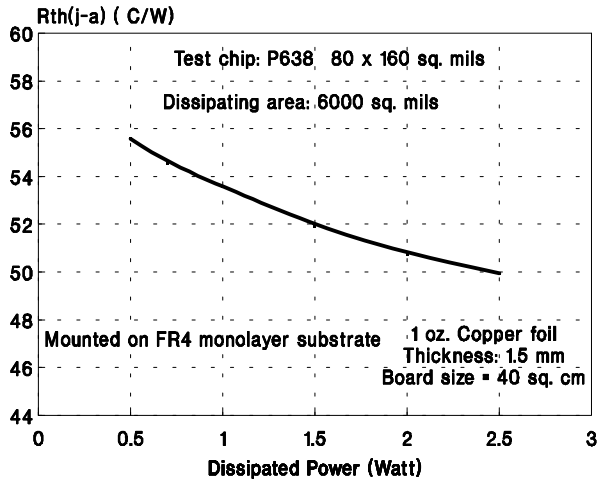


Figure 13. PowerSO20: with external heatsink

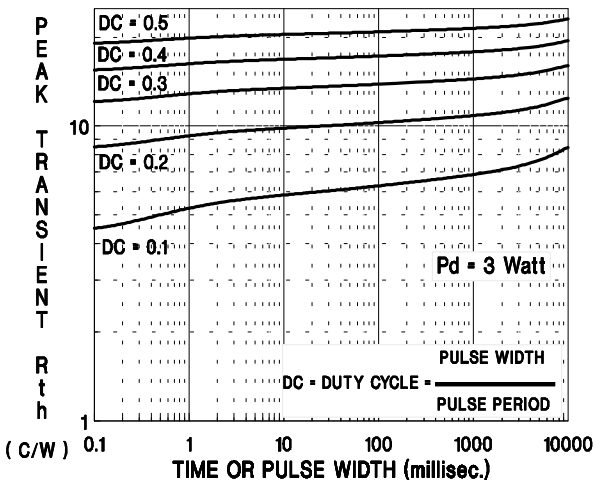
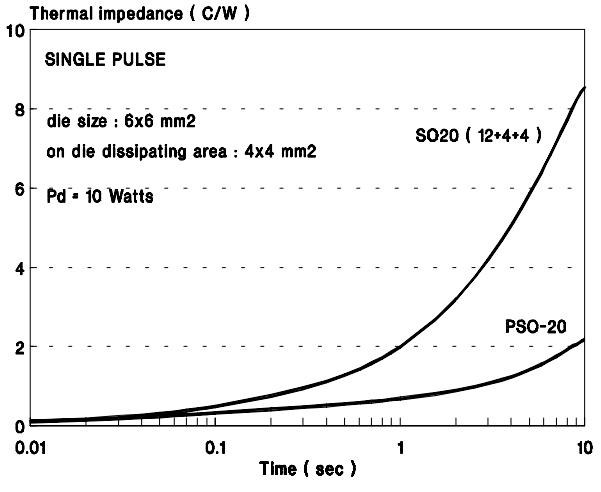


Figure 14. Thermal impedance of PowerSO20 and standard SO20



8 Package information

In order to meet environmental requirements, ST offers these devices in different grades of ECOPACK® packages, depending on their level of environmental compliance. ECOPACK® specifications, grade definitions and product status are available at: *www.st.com*. ECOPACK® is an ST trademark.

8.1 PowerSO20 package information

Figure 15. PowerSO20 package outline

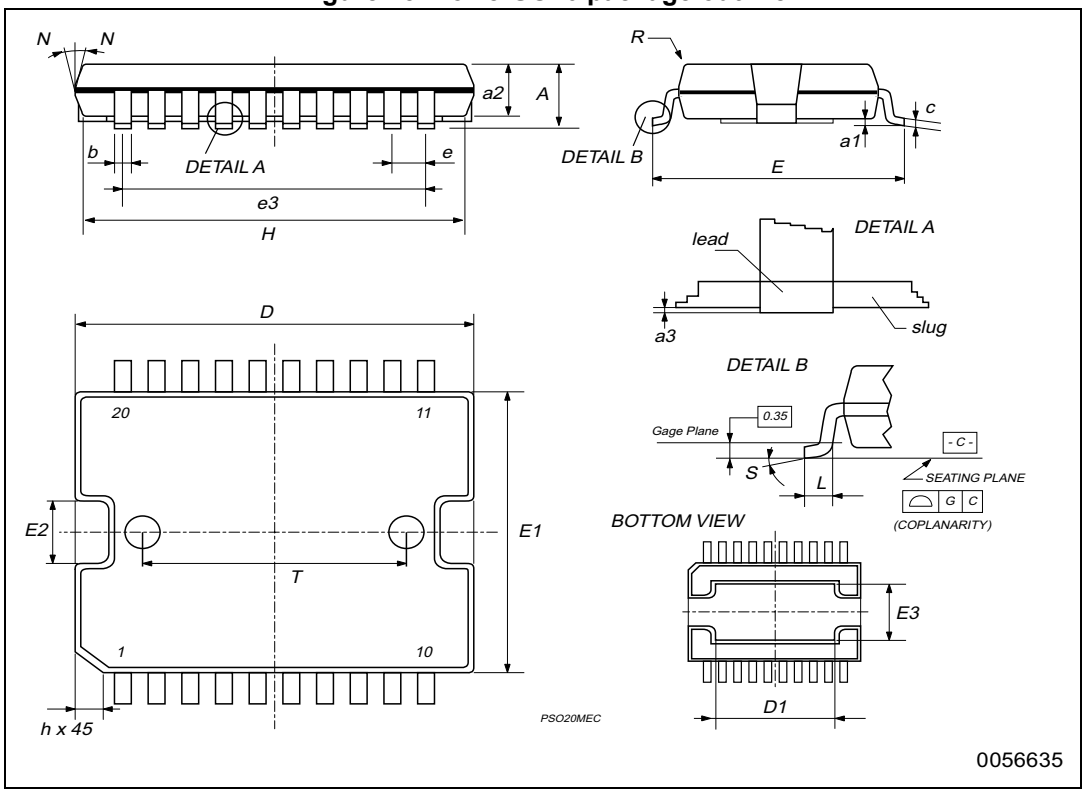


Table 7. PowerSO20 package mechanical data

Symbol	Dimensions (mm)		
	Min.	Typ.	Max.
A	-	-	3.6
a1	0.1	-	0.3
a2	-	-	3.3
a3	0	-	0.1
b	0.4	-	0.53
c	0.23	-	0.32
D(1)	15.8	-	16
D1	9.4	-	9.8
E	13.9	-	14.5
e	-	1.27	-
e3	-	11.43	-
E1(1)	10.9	-	11.1
E2	-	-	2.9
E3	5.8	-	6.2
G	0	-	0.1
H	15.5	-	15.9
h	-	-	1.1
L	0.8	-	1.1
N	8° (typ.)		
S	8° (max.)		
T	-	10	-

1. “D” and “E1” do not include mold flash or protrusions.
   - Mold flash or protrusions shall not exceed 0.15 mm (0.006").

9 Revision history

Table 8. Document revision history

Date	Revision	Changes
15-Nov-2011	10	Updated Features in coverpage and *Table 4*.
15-Mar-2017	11	Removed PowerDIP 20-pin package and all references throughout document. Updated *Figure 1 on page 3* (replaced by new figure). Added note *1.* below *Table 7 on page 13*. Minor modifications throughout document.

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